

Patent Search and Prior Art (India)

A search of Indian patents and published applications for the various Aetheris OS features identified only a few relevant filings. Notably, **Indian Patent IN530474 (granted)** covers a “*blockchain-based micro-blogging*” system ¹. This IIT Madras invention runs a decentralized microblog (each mobile device is a blockchain node) and performs consensus-based content validation ² ³. We also found two Indian patents involving AI and operating systems, though they address very different problems. **IN239319** claims a “*proactive user interface*” with an emotional AI agent layered over an OS ⁴, and **IN228347** claims an AI-driven object-clustering display system ⁵. These patents use AI components (e.g. pattern-detection in IN239319) but focus on UI features, not on any novel AI algorithms or hardware integration ⁶. Aside from these, no Indian patent or published application was found for XR/AR operating systems, DAO governance, privacy-preserving (e.g. homomorphic) AI, deterministic AI, polyglot runtimes, or a native decentralized OS in general. (For example, a VMware application (IN201941003120) on OS update management exists ⁷, but it merely handles patch deployment and is unrelated to Aetheris OS’s novel features.)

No Indian patent was found specifically on Extended Reality (XR) OS integration or on a DAO-based OS. Similarly, we found **no Indian filings** for AI-scheduling OS kernels or for built-in multi-language runtimes. In contrast, international patents like WO2017/187397A1 (an IoT blockchain OS) ⁸ and US patents on “blockchain operating systems” exist, but they have no Indian counterparts. In sum, Indian filings largely cover niche applications (UI tools, microblogging) and not a unified intelligent blockchain OS.

Identified Relevant Patents and Applications

- **IN530474 (granted)** – “*Blockchain-based micro-blogging*” ¹. A fully decentralized social platform where each mobile device is a blockchain node that runs a microblog client with REST API and consensus (e.g. Stellar) ⁹ ¹⁰. User content is validated by federated ML and blockchain consensus before posting ³. *Overlap:* Aetheris OS’s “native decentralized social networking layer” partially overlaps conceptually, but IN530474 is an application-layer microblog (not an OS). It teaches decentralized content storage and consensus rewards. Aetheris OS must **avoid replicating this specific microblog workflow** (e.g. Stellar consensus and mobile-node design) if it provides similar features, for instance by using different consensus or integrating social networking as an OS service rather than a standalone app.
- **IN239319 (granted)** – “*Proactive user interface with AI agent*” ⁴. Claims an OS-based UI that uses an AI framework to detect user patterns and suggest function changes, with an emotional agent. *Overlap:* Minimal. This patent is about an AI-enhanced GUI, not about OS architecture. Aetheris OS’s AI features are at the system level (e.g. resource scheduling, privacy) rather than a user-interface layer. Thus Aetheris OS would **not infringe IN239319**, as it does not claim to alter UI functions with an “emotional agent”.
- **IN228347 (granted)** – “*AI-based object clustering display*” ⁵. Claims a display system that clusters related objects and an AI component to select which to display. *Overlap:* None with Aetheris OS. This

patent concerns data visualization. Aetheris OS may manage data and AI, but it would not infringe a claim about arranging GUI objects in clusters.

- **IN201941003120 (published application)** – “Operating System Update Management” (VMware). Titled “Operating System Update Management”, this Indian application (priority Jan 2019) relates to scheduling OS patches in enterprise settings ⁷. *Overlap*: Irrelevant. Aetheris OS’s core innovations are unrelated to update scheduling.
- **Other patents found (non-Indian)**: We note e.g. WO2017/187397A1 (“blockchain IoT OS”) ⁸ and US11689513B2 (“blockchain operating system”) ¹¹. These teach a blockchain-linked OS concept, but no corresponding Indian filings exist. They confirm that *generic blockchain-OS ideas* have been explored elsewhere. Aetheris OS’s integration of blockchain and OS could draw on these concepts, but since they are not Indian-patented, they pose no direct infringement risk in India.

Conflict and Infringement Analysis

- **IN530474 (blockchain microblogging)**: Legally, this patent’s scope is limited to a blockchain-based microblogging platform. Aetheris OS would include a social layer, but to avoid infringement it should emphasize **system-level integration** rather than mimicking the patented application’s method. For example, IN530474 uses a specific Stellar-consensus and mobile-node design ⁹; Aetheris OS can use a different consensus or implement social networking as a kernel-level service. Technically, there is little overlap: IN530474 claims a federated microblog architecture, whereas Aetheris OS claims an OS architecture. **No infringement** is expected as long as Aetheris does not copy the exact content moderation or node-based reward mechanism of IN530474.
- **IN239319 & IN228347 (AI UI patents)**: These patents focus narrowly on UI behavior and display. Aetheris OS’s AI modules (for resource management, privacy, etc.) operate entirely differently. There is **no technical overlap**: Aetheris does not claim an “emotional agent” or the specific object-clustering UI. Therefore, Aetheris OS will **not infringe** these patents. Legally, these patents illustrate that Indian examiners will enforce CRI (computer-related invention) guidelines; they teach that claims must highlight hardware or system effects ¹² ⁶.
- **Legal Guidelines (CRI §3(k))**: The TTConsultants analysis ¹² warns that purely software claims are not allowable. To ensure patentability, Aetheris OS must emphasize **hardware integration and technical effect**. For example, claims should mention hardware components (CPUs, sensors, network interfaces) and practical improvements (security, speed, scalability) ¹². The prior patents exploited narrow “AI” features only in system functionality ⁶; we must claim the *system as a whole*, not algorithms per se.
- **Infringement Risk**: Given the above, Aetheris OS as proposed should **not infringe any of the identified Indian patents**. The known patents cover very specific applications (UI or microblogging). Aetheris OS’s architecture (a multi-module OS combining XR, AI, blockchain, etc.) is broader. By drafting the claims to stress OS-level integration and including hardware components (e.g. trusted execution environment, blockchain modules, XR sensors), we can both align with 3(k) and avoid existing patent scopes.

Gaps and Patentability Opportunities

- **Narrow Scopes of Prior Art:** The existing patents (IN530474, IN239319, IN228347) are very limited in scope ¹ ⁴ . They leave ample **gaps** to claim Aetheris OS's innovations. For example, none of these patents address: integrated AI for deterministic resource scheduling, or an OS managing federated learning with privacy. We can exploit these gaps by emphasizing such novel features.
- **Absence of XR and Web3 in Indian Filings:** No Indian patent covers XR integration or Web3 in an OS. Thus Aetheris OS's XR framework and native blockchain support are *novel in India*. Claims can stress the unique combination of XR + blockchain at the OS level.
- **DAO Governance:** No Indian application was found on DAO-based system governance. We can claim, e.g., a **tokenized governance layer** that manages OS permissions and updates, leveraging smart contracts. This is a strong novelty point given the lack of prior art.
- **Privacy-Preserving AI:** Techniques like homomorphic encryption or zero-knowledge protocols integrated into the OS are not found in patents. We should claim the “privacy engine” explicitly, highlighting e.g. secure enclaves or encrypted analytics.
- **Polyglot Runtime Environment:** Indian patents have not addressed multi-language runtimes. We can claim an OS that includes a language-agnostic execution platform (for example, a WebAssembly or container subsystem that allows seamless execution of multiple language modules).
- **Combined Technical Effect:** Perhaps most importantly, no single patent claims the **overall synergy** of these features. Aetheris OS can argue inventive step on the basis of this combination: e.g. “by integrating AI, blockchain, XR, DAO, and multi-language runtime in one OS, system latency is reduced, security is enhanced, and developer flexibility is improved” – effects not achieved by prior art.

By drafting claims around these novel combinations and embedding hardware/firmware details (per CRI guidelines ¹²), the Aetheris OS application can be made both broad and robust. We avoid the prior art by focusing on *operating system innovations* (not just UI or apps) and by stressing novel subsystems that the identified patents do **not** cover.

Draft Indian Patent Application for Aetheris OS

Title: “AETHERSIS: An AI-Integrated Decentralized Operating System with Extended Reality, Blockchain, and Polyglot Runtime”

Abstract: Aetheris OS is a computing platform that unifies extended-reality (XR) interfaces, artificial intelligence (AI) computing, blockchain/Web3 integration, decentralized cloud/edge computing, and on-device social networking within a single operating system. In one embodiment, the OS includes an AI-driven resource manager, an immersive XR user interface engine, a built-in distributed ledger module, and a governance module based on a decentralized autonomous organization (DAO). A polyglot runtime supports multiple programming languages via a common virtual machine. Privacy is preserved through on-device

encryption and federated learning, and deterministic AI ensures reproducible outcomes. The OS dynamically offloads tasks among network peers for edge computing. By combining these modules, Aetheris OS enables novel applications (e.g. secure metaverse services) not possible with conventional systems. This integrated architecture provides a technical solution for high-security, high-scalability computing across heterogeneous devices, demonstrating novelty and industrial applicability.

Field of the Invention: This invention relates generally to computer operating systems and, more particularly, to an operating system that natively integrates extended reality (XR), artificial intelligence (AI), distributed blockchain/Web3 technology, decentralized cloud/edge computing, and multi-language execution. The invention further relates to system-level governance via decentralized autonomous organization (DAO) concepts and privacy-enhancing AI within the OS.

Background: Modern computing platforms (smartphones, PCs, and servers) are increasingly expected to support immersive XR experiences, AI-powered services, and secure, decentralized applications. However, existing operating systems (e.g. Android, Windows) treat these as separate layers: VR/AR, AI, and blockchain are typically handled by standalone applications or middleware, not the OS itself. This leads to inefficiencies in resource sharing, security vulnerabilities, and high development overhead. For example, current systems do not natively synchronize a blockchain ledger with OS processes, nor do they provide built-in XR rendering coupled with system-wide AI. Similarly, governance of system updates and configurations is centrally controlled, not leveraging token-based consensus. There is a need for an OS that provides **intrinsic support** for XR, AI, decentralized computing and governance, polyglot execution, and privacy-preserving data processing, all at the kernel or system-service level.

Summary of the Invention: The invention provides **Aetheris OS**, a novel operating system architecture that integrates the following modules: (1) an XR Interface Module, which handles VR/AR sensor data and renders immersive interfaces; (2) an AI Processing Module, employing deterministic machine-learning models for tasks like resource scheduling, vision, and natural language; (3) a Blockchain/Web3 Module, maintaining a distributed ledger within the OS for secure transaction logging and decentralized services; (4) a Decentralized Computing Module, enabling peer-to-peer task offloading to other Aetheris nodes (supporting edge and cloud cooperation); (5) a DAO Governance Module, in which system policies and updates are managed by on-chain voting among participants; (6) a Privacy Module, which applies homomorphic encryption and federated learning so that sensitive data remains encrypted and never leaves the device; and (7) a Polyglot Runtime Module, which executes applications written in diverse languages (via a unified VM or container, e.g. WebAssembly) without recompilation. Aetheris OS also includes a Social Networking Service Layer at the OS level, allowing peer-to-peer content exchange without centralized servers. These components operate cooperatively: for instance, the AI Module may adapt resource allocations based on user gaze from the XR Module; the Blockchain Module may record verified sensor data; and the DAO Module may authorize execution of applications. In combination, Aetheris OS achieves improved security (via blockchain and encryption), flexibility (via multi-language support), and performance (via intelligent scheduling and edge collaboration). This architecture is novel and non-obvious, as no single prior system provides all these features integrated at the OS level.

Brief Description of the Drawings:

- FIG. 1 illustrates a high-level block diagram of a computing device running Aetheris OS, showing major modules (XR Interface, AI Processor, Blockchain Ledger, etc.) and their interactions.
- FIG. 2 is a flowchart of decentralized task scheduling: devices form an ad-hoc edge network, broadcast compute tasks to peers, and aggregate results via the blockchain ledger.

- FIG. 3 shows an example of DAO-based governance: stakeholders vote on a proposed OS update through smart contracts, and upon consensus the update is automatically verified and installed.

Detailed Description of the Invention: Referring to FIG. 1, Aetheris OS runs on a **multicore computing device** (e.g. a smartphone or AR headset) with hardware components such as GPU/TPU accelerators 102, XR sensors (camera, lidar, IMU) 104, and a secure enclave 106. The **XR Interface Module** 120 processes raw sensor inputs 104 (e.g. depth data, position) to render immersive displays or audio. It provides a unified spatial computing framework for applications. The **AI Processing Module** 130 includes deterministic neural-network engines and on-chip AI accelerators (102) to perform tasks such as object recognition in video, language understanding, and predictive caching. All AI models have fixed initialization and use on-device datasets, ensuring deterministic behavior. The AI Module communicates with both the XR Module (e.g. to label visual objects) and system services (e.g. resource manager).

The **Blockchain/Web3 Module** 140 runs a light client of a permissioned blockchain. It stores system state (OS configuration, logs of transactions, smart contracts) in a distributed ledger replicated among devices. For example, OS updates, security keys, and inter-device messages are committed to this blockchain for tamper-evidence. A consensus engine within 140 uses a Proof-of-Stake or Practical Byzantine Fault Tolerance (PBFT) algorithm suited to resource-constrained devices. In one embodiment, GPU/TPU 102 is used to accelerate cryptographic hashing.

The **Decentralized Computing Module** 150 enables **fog/edge computing**. When the device is connected to a local mesh network, it can offload heavy tasks (e.g. AI inference or XR rendering slices) to peer devices running Aetheris OS, and receive results back. This is coordinated via a service queue on the blockchain ledger 140 (which logs available tasks and results). Each device advertises its capabilities and workload to others; a distributed scheduler in 150 (powered by the AI Module) distributes tasks to peers to optimize latency and energy use.

The **DAO Governance Module** 160 governs system-wide policies. When a software or configuration change is proposed (e.g. a new kernel version), an on-chain ballot is created using smart contract logic. Token-holding participants (which may include device owners or even autonomous agents) vote on the proposal. A consensus threshold (e.g. 66%) is required for approval. Once passed, the OS update is automatically fetched, verified by cryptographic signatures, and applied. In this way, upgrades and security rules are administered by decentralized agreement rather than a central authority.

The **Privacy Module** 170 ensures user data never leaves the device in raw form. For example, when performing AI on sensitive data (health readings, biometric scans), the data is first run through a **homomorphic encryption engine** and only encrypted statistics (or masked gradient updates) are shared. Federated learning protocols are used for model updates: each device trains on its local data, and only encrypted model deltas are combined on-chain (via the blockchain module) to update a global model. Aetheris OS also uses **secure enclaves 106** for trusted execution of critical code.

The **Polyglot Runtime Module** 180 contains an interpreter/VM (e.g. WebAssembly runtime or language VM) and container manager that can execute applications written in any supported language (C/C++, Rust, JavaScript, etc.) without OS modifications. This is achieved by packaging apps in language-agnostic containers that interface with OS services through a universal API. The OS kernel 190 mediates all resource access.

Finally, the **Social Networking Layer** 200 provides APIs for peer-to-peer communication and content sharing. Rather than relying on external servers, Aetheris OS devices exchange messages and media directly via encrypted channels. Access control lists and identity verification are managed via blockchain 140. Content moderation (e.g. filtering hate speech) can be done locally by AI 130 or via a federated consensus.

Example Use-Case: A user dons XR glasses running Aetheris OS. The XR Interface 120 displays an augmented workplace. Meanwhile, a local Aetheris “compute cluster” of devices processes a machine vision task using Decentralized Computing 150. The AI Module 130 dynamically reallocates CPU/GPU resources as needed. The user performs a transaction (e.g. medical data sharing) through a blockchain-based app, with records stored on 140. Later, the device automatically applies a new security update after the community (160) votes it safe. Throughout, private data never leaves the device thanks to module 170. This scenario illustrates how Aetheris OS’s integrated design enables seamless, secure, and intelligent operation.

Advantages: Aetheris OS’s unified architecture yields several benefits: improved performance by co-locating XR and AI, enhanced security via an on-chain ledger and enclave, flexibility via multi-language support, and resilience through decentralized task distribution. These combined features are not taught or suggested by any single prior patent ¹ ⁴, ensuring the claimed invention is novel and non-obvious.

Claims

1. **(Independent)** A computing system comprising:
a multicore processor and graphics/AI accelerators;
memory storing an operating system; and
program instructions implementing an **operating system** that includes:
 2. an *extended reality interface module* that processes multi-sensor XR data (camera, lidar, IMU) to render immersive visuals and audio;
 3. an *artificial intelligence module* that executes deterministic machine-learning models for tasks including resource management and sensor data analysis;
 4. a *distributed ledger module* that maintains a blockchain-based log of system state and transactions, the blockchain replicated across multiple devices;
 5. a *decentralized edge computing module* that offloads computational tasks to peer devices via a peer-to-peer network using the blockchain for coordination;
 6. a *governance module* that manages system updates and policies through on-chain voting (decentralized autonomous organization) by stakeholders;
 7. a *privacy module* that applies homomorphic encryption and federated learning so that sensitive data is encrypted before sharing with any other device;
 8. a *polyglot runtime module* that provides language-agnostic execution support (e.g. via a WebAssembly or VM environment) for applications written in various programming languages; and
 9. a *social networking service layer* that enables peer-to-peer content exchange without centralized servers;
wherein the modules cooperate to improve security and performance (for example, the AI module adjusts resource allocation based on XR sensor inputs, and the blockchain module records verified events from the XR interface), thereby providing an integrated XR/AI/blockchain operating environment not found in the prior art.

10. The system of claim 1, wherein the extended reality interface module is coupled to a wearable headset and one or more cameras, and wherein the graphics accelerator processes virtual object textures in real time.
11. The system of claim 1, wherein the artificial intelligence module utilizes a fixed-weight neural network and integrated tensor-processing hardware to ensure determinism, and wherein it adjusts task scheduling and power allocation based on workload predictions.
12. The system of claim 1, wherein the distributed ledger module implements a proof-of-stake consensus and cryptographic hashing accelerated by a GPU, and stores smart contracts that govern inter-device services.
13. The system of claim 1, wherein the decentralized edge computing module dynamically assigns subtasks of a workload to other peer devices running the operating system, and aggregates results via the blockchain.
14. The system of claim 1, wherein the governance module encodes proposed system configuration changes in blockchain transactions and requires a configurable quorum of token-weighted votes for approval, upon which the operating system automatically fetches and verifies signed updates.
15. The system of claim 1, wherein the privacy module encloses private inputs in a secure enclave and participates in federated learning by sharing only encrypted model updates via the blockchain.
16. The system of claim 1, wherein the polyglot runtime module supports execution of applications written in native code (e.g. C/C++), managed code (e.g. Java), and scripting languages (e.g. JavaScript) by converting them to a common intermediate representation.
17. The system of claim 1, wherein the social networking layer provides end-to-end encrypted messaging and content sharing directly between devices, and moderates content using on-device AI classifiers.
18. (Dependent) A method of operating a computing device comprising: initializing the modules of claim 1; receiving XR sensor streams and rendering them; performing machine-learning inference on incoming data; creating or verifying blockchain transactions for system events; offloading a portion of a computing task to a peer device and receiving the result; conducting a token-based vote on a proposed update; and applying privacy transformations to data in accordance with claim 1.
19. The method of claim 10, further including: instantiating a secure multiparty computation or zero-knowledge proof to validate a transaction without revealing underlying data.
20. (Dependent) The system of claim 1, wherein the operating system is deployed on a network of IoT devices and desktop PCs, and the modules communicate over wireless links using a consensus-backed protocol.

Examples: In one example, three devices running Aetheris OS form an ad-hoc computing cluster. Device A detects an XR object and generates a heavy vision workload. Using claim 5's edge module, A dispatches sub-

tasks to B and C; each returns its analysis, which is recorded on the blockchain (claim 4) and presented to the user through A's XR interface. Meanwhile, any configuration change (e.g. new XR drivers) is proposed as a transaction; devices B and C vote in the DAO (claim 6), reaching consensus so that A automatically installs the update. Throughout, user biometrics never leave any device in the clear (claim 7).

This invention, as claimed, is novel, involves an inventive step, and is industrially applicable to any computing environment where AI, XR, and decentralized trust are needed. The foregoing detailed description and examples are illustrative; the claims define the full scope of the invention.

1 2 3 9 10 IDF No 2159 A System For Blockchain-based Micro-blogging And A Method Thereof – Industrial Consultancy & Sponsored Research (IC&SR) , IIT Madras
<https://ipm.icsr.in/ipm/block-chain/idf-no-2159-a-system-for-blockchain-based-micro-blogging-and-a-method-thereof>

4 5 6 12 Can Artificial Intelligence software be patented in India? - TT CONSULTANTS
<https://ttconsultants.com/can-artificial-intelligence-software-be-patented-in-india/>

7 US20200241868A1 - Operating system update management - Google Patents
<https://patents.google.com/patent/US20200241868A1/en>

8 WO2017187397A1 - Operating system for blockchain iot devices - Google Patents
<https://patents.google.com/patent/WO2017187397A1/en>

11 US11689513B2 - Blockchain operating system - Google Patents
<https://patents.google.com/patent/US11689513B2/en>